

# A weighted tensor-grid subcase of the Schatten linear-KS question

Candidate partial-result packet for arXiv:1501.05213

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**Status.** `candidate partial result -- likely valid`. This note does not solve the full question whether the Schatten trace class  $S_1$  is a linear KS space or a KS metric space.

## 1 Source question

Naor and Schechtman ask whether  $S_1$  satisfies their original linear Kwapien–Schütt inequality and, after Definition 1.4, whether  $S_1$  is a KS metric space or even a linear KS space [1]. The relevant passage is on pages 3–4 of the arXiv PDF.

For an array  $(z_{jk})_{j,k=1}^n$  in a Banach space  $Z$ , write

$$\begin{aligned} L(z) &= \frac{1}{n} \sum_{j=1}^n \mathbb{E}_{\delta} \left\| \sum_{k=1}^n \delta_k z_{jk} \right\|_Z, \\ R_{\text{map}}(z) &= \mathbb{E}_{K \in [n]^n} \mathbb{E}_{\varepsilon} \left\| \sum_{j=1}^n \varepsilon_j z_{j, K_j} \right\|_Z, \\ R_{\text{perm}}(z) &= \mathbb{E}_{\pi \in S_n} \mathbb{E}_{\varepsilon} \left\| \sum_{j=1}^n \varepsilon_j z_{j, \pi(j)} \right\|_Z. \end{aligned}$$

The modified linear-KS question asks whether  $L(z) \leq C R_{\text{map}}(z)$  for all  $S_1$ -valued arrays with a universal  $C$ . The earlier inequality uses  $R_{\text{perm}}$  instead.

**Definition 1.4** (Linear KS space). Say that a Banach space  $(Z, \|\cdot\|_Z)$  is a linear KS space if there exists  $C = C(X) \in (0, \infty)$  such that for every  $n \in \mathbb{N}$  and every  $\{z_{jk}\}_{j,k \in \{1, \dots, n\}} \subseteq Z$  we have

$$\frac{1}{n} \sum_{j=1}^n \sum_{\varepsilon \in \{-1, 1\}^n} \left\| \sum_{k=1}^n \varepsilon_k z_{jk} \right\|_Z \leq \frac{C}{n^n} \sum_{k \in \{1, \dots, n\}^n} \sum_{\varepsilon \in \{-1, 1\}^n} \left\| \sum_{j=1}^n \varepsilon_j z_{jk_j} \right\|_Z. \quad (10)$$

The difference between (10) and (8) is that we replace the averaging over all permutations  $\pi \in S_n$  by averaging over all mappings  $\pi : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ . We shall see below that  $L_1$  is a linear KS space. The same reasoning that leads to (9) now leads us to consider the following new bi-Lipschitz invariant for metric spaces.

**Definition 1.5** (KS metric space). Say that a metric space  $(X, d_X)$  is a KS space if there exists  $C = C(X) \in (0, \infty)$  such that for every  $n \in 2\mathbb{N}$  and every  $f : M_n(\mathbb{F}_2) \rightarrow X$  we have

$$\frac{1}{n} \sum_{j=1}^n \sum_{x \in M_n(\mathbb{F}_2)} d_X \left( f \left( x + \sum_{k=1}^n e_{jk} \right), f(x) \right) \leq \frac{C}{n^n} \sum_{k \in \{1, \dots, n\}^n} \sum_{x \in M_n(\mathbb{F}_2)} d_X \left( f \left( x + \sum_{j=1}^n e_{jk_j} \right), f(x) \right). \quad (11)$$

*Remark 1.6.* The reason why in Definition 1.5 we require (11) to hold true only when  $n \in \mathbb{N}$  is even is that no non-singleton metric space  $(X, d_X)$  satisfies (11) when  $n \geq 3$  is an odd integer. Indeed, suppose that  $a, b \in X$  are distinct and that  $n \geq 3$  is an odd integer. For  $x \in M_n(\mathbb{F}_2)$  write  $\sigma(x) = \sum_{j=1}^{n-1} \sum_{k=1}^n x_{jk} \in \mathbb{F}_2$ . Define  $f : M_n(\mathbb{F}_2) \rightarrow X$  by setting  $f(x) = a$  if  $\sigma(x) = 0$  and  $f(x) = b$  if  $\sigma(x) = 1$ . For every  $j \in \{1, \dots, n-1\}$  and  $x \in M_n(\mathbb{F}_2)$  we have  $\sigma(x + \sum_{k=1}^n e_{jk}) = \sigma(x) + n \neq \sigma(x)$ , since  $n$  is odd. Consequently the left hand side of (11) is nonzero (since  $a$  and  $b$  are distinct). But, for every  $x \in M_n(\mathbb{F}_2)$  and  $k \in \{1, \dots, n\}^n$  we have  $\sigma(x + \sum_{j=1}^n e_{jk_j}) = \sigma(x) + n - 1 = \sigma(x)$ , since  $n$  is odd. Consequently the right hand side of (11) vanishes. This parity issue is of minor significance: in Remark 2.3 below we describe an inequality that is slightly more complicated than (11) but make sense for every  $n \in \mathbb{N}$  and in any metric space, and we show that it holds true for  $L_1$ -valued functions. This variant has the same nonembeddability consequences as (11), albeit yielding distortion lower bounds that are weaker by a constant factor.

The following theorem is the main result of the present article. We shall soon see, in Section 1.2 below, how it quickly implies Theorem 1.1 (and more).

**Theorem 1.7.**  $L_1$  is a KS space. Namely, for all  $n \in 2\mathbb{N}$  and every  $f : M_n(\mathbb{F}_2) \rightarrow L_1$  we have

$$\sum_{j=1}^n \sum_{x \in M_n(\mathbb{F}_2)} \left\| f \left( x + \sum_{k=1}^n e_{jk} \right) - f(x) \right\|_1 \leq \frac{2n}{n^n - (n-2)^n} \sum_{k \in \{1, \dots, n\}^n} \sum_{x \in M_n(\mathbb{F}_2)} \left\| f \left( x + \sum_{j=1}^n e_{jk_j} \right) - f(x) \right\|_1.$$

For every fixed  $n \in 2\mathbb{N}$ , the factor  $2n/(n^n - (n-2)^n)$  above cannot be improved. So,  $L_1$  satisfies (11) for every  $n \in 2\mathbb{N}$  with

$$C = \sup_{n \in 2\mathbb{N}} \frac{2}{1 - \left(1 - \frac{2}{n}\right)^n} = \frac{2e^2}{e^2 - 1},$$

and this value of  $C$  cannot be improved.

Note that if  $(Z, \|\cdot\|_Z)$  is a Banach space and  $\{z_{jk}\}_{j,k \in \{1, \dots, n\}} \subseteq Z$  then by considering the mapping  $f : M_n(\mathbb{F}_2) \rightarrow Z$  given by  $f(x) = \sum_{j=1}^n \sum_{k=1}^n (-1)^{x_{jk}} z_{jk}$  we see that if  $Z$  is a KS space as a metric space then it is also a linear KS space (with the same constant  $C$ ). We do not know whether or not the converse holds true, i.e., we ask the following interesting open question: if a Banach space  $(Z, \|\cdot\|_Z)$  is a linear KS space then is it also a KS space as a metric space? Understanding which Banach spaces are linear KS spaces is a wide-open research direction. In particular, we ask whether the Schatten trace class  $S_1$  is a KS space as a metric space, or even whether it is a linear KS space.

## 2 Idea of the proof

The theorem handles the canonical noncommutative tensor grid  $u_j v_k^*$  with arbitrary scalar weights. Within one row, a signed sum is rank one, so its trace norm is simply the Euclidean norm of the coefficient row. A permutation selects orthogonal initial and terminal vectors, so its trace norm is the sum of the selected absolute weights.

For an arbitrary mapping  $K : [n] \rightarrow [n]$ , collisions are the only new feature. Grouping the selected rank-one operators by their common terminal vector  $v_k$  gives mutually orthogonal column blocks. Their Euclidean norms are exactly the singular values. In each block we retain only the largest selected coefficient. A decreasing rearrangement and a first-success computation for independent Bernoulli variables of mean  $1/n$  then supplies the uniform factor  $(1 - 1/n)^{n-1} \geq e^{-1}$ . This proves the modified inequality with constant  $e$ . The full  $S_1$  problem remains open because a general array has no fixed two-sided orthogonal tensor grid.

## 3 Partial theorem

**Theorem 1** (weighted tensor grids). *Let  $H$  be a real or complex Hilbert space, let  $(u_j)_{j=1}^n$  and  $(v_k)_{k=1}^n$  be orthonormal families in  $H$ , and let  $(a_{jk}) \in \mathbb{F}^{n \times n}$ . Define*

$$z_{jk} = a_{jk} u_j v_k^* \in S_1(H).$$

Then

$$L(z) \leq R_{\text{perm}}(z) \quad \text{and} \quad L(z) \leq e R_{\text{map}}(z).$$

Thus source equations (8) and (10) hold uniformly on every weighted tensor-grid array, with constants 1 and  $e$ , respectively.

*Proof.* For fixed  $j$  and every sign vector  $\delta$ ,

$$\sum_{k=1}^n \delta_k z_{jk} = u_j \left( \sum_{k=1}^n \overline{\delta_k a_{jk}} v_k \right)^*.$$

This operator has rank at most one and trace norm  $(\sum_k |a_{jk}|^2)^{1/2}$ . Hence

$$L(z) = \frac{1}{n} \sum_{j=1}^n \left( \sum_{k=1}^n |a_{jk}|^2 \right)^{1/2}. \quad (1)$$

If  $\pi \in S_n$ , then the initial vectors  $v_{\pi(j)}$  and terminal vectors  $u_j$  are both orthonormal. Therefore the singular values of  $\sum_j \varepsilon_j a_{j,\pi(j)} u_j v_{\pi(j)}^*$  are  $|a_{j,\pi(j)}|$ , independently of  $\varepsilon$ . It follows that

$$R_{\text{perm}}(z) = \mathbb{E}_{\pi} \sum_{j=1}^n |a_{j,\pi(j)}| = \frac{1}{n} \sum_{j,k=1}^n |a_{jk}|.$$

Together with the rowwise inequality  $\|(a_{jk})_k\|_2 \leq \|(a_{jk})_k\|_1$ , this and (1) prove  $L(z) \leq R_{\text{perm}}(z)$ .

Now fix  $K = (K_1, \dots, K_n) \in [n]^n$ . Grouping by the selected terminal index gives

$$\sum_{j=1}^n \varepsilon_j z_{j,K_j} = \sum_{k=1}^n \left( \sum_{j:K_j=k} \varepsilon_j a_{jk} u_j \right) v_k^*.$$

The vectors in parentheses are pairwise orthogonal because their sets of  $u_j$  coordinates are disjoint, and the  $v_k$  are orthonormal. Consequently the nonzero singular values are

$$\left( \sum_{j:K_j=k} |a_{jk}|^2 \right)^{1/2}, \quad k \in [n],$$

and hence

$$R_{\text{map}}(z) = \sum_{k=1}^n \mathbb{E}_K \left( \sum_{j=1}^n |a_{jk}|^2 \mathbf{1}_{\{K_j=k\}} \right)^{1/2}. \quad (2)$$

Fix a column  $k$  and rearrange its absolute weights as  $b_1 \geq \dots \geq b_n \geq 0$ . The indicators  $B_r = \mathbf{1}_{\{K_{j_r}=k\}}$  are independent Bernoulli variables with mean  $1/n$ . Since the Euclidean norm dominates the maximum,

$$\begin{aligned} \mathbb{E} \left( \sum_{r=1}^n b_r^2 B_r \right)^{1/2} &\geq \mathbb{E} \max_{1 \leq r \leq n} b_r B_r \\ &= \frac{1}{n} \sum_{r=1}^n b_r \left( 1 - \frac{1}{n} \right)^{r-1} \\ &\geq \frac{1}{n} \left( 1 - \frac{1}{n} \right)^{n-1} \sum_{r=1}^n b_r \geq \frac{1}{en} \sum_{r=1}^n b_r. \end{aligned}$$

For  $n = 1$  the same conclusion is immediate. Summing this estimate over  $k$  in (2) yields

$$R_{\text{map}}(z) \geq \frac{1}{en} \sum_{j,k=1}^n |a_{jk}| \geq \frac{1}{e} L(z),$$

where the final inequality again uses the rowwise  $\ell_2$ - $\ell_1$  comparison and (1). This proves the theorem.  $\square$

## 4 Verification and limitations

The proof has three checkable inputs: the rank-one row formula, the orthogonal column-block singular-value formula, and the Bernoulli first-hit identity. The accompanying verifier script enumerates all selectors on randomized weighted arrays for  $1 \leq n \leq 5$  and directly checks the singular-value formula using matrix SVDs. These computations are regression checks only; the proof above is exact.

The result does not cover arrays  $z_{jk}$  whose left or right singular directions depend on both indices. It does not imply that  $S_1$  is a linear KS space, and it gives no nonlinear metric KS inequality. No optimality claim is made for the constant  $e$ .

## 5 Novelty-check status

The run's lightweight indexes had no record for arXiv:1501.05213 or this subcase. A bounded exact-phrase, arXiv, and forward-citation search on 19 July 2026 found no claimed resolution of the  $S_1$  linear-KS question or this weighted tensor-grid statement. The later Naor-Schechtman material on Schatten embeddings uses the distinct upper-property- $\alpha$  inequality. This bounded search does not certify novelty.

## 6 Human-review recommendation

Check equation (2) first. Once its singular-value identification is accepted, the probability estimate and both conclusions are elementary. If retained, this packet should remain classified as a narrow partial result; the source’s general linear and metric questions are untouched.

## References

- [1] A. Naor and G. Schechtman, *Pythagorean powers of hypercubes*, Ann. Inst. Fourier (Grenoble) 66 (2016), no. 3, 1093–1116; arXiv:1501.05213; doi:10.5802/aif.3032.